1a.)

$$A_1 = \begin{pmatrix} 2 & 1 \\ -1 & 2 \end{pmatrix}$$

$\frac{1}{2}R1 + R2 \to R2$

$$\begin{pmatrix} 2(1/2) & 1(1/2) \\ -1 & 2 \end{pmatrix}$$

$$\begin{pmatrix} 2 & 1 \\ 0 & 5/2 \end{pmatrix} = U$$

L-matrix:

$$A_1 = \begin{pmatrix} 2 & 1 \\ -1 & 2 \end{pmatrix} \quad -2R1 + R2 \to R1$$

$$\begin{pmatrix} (2)(-2) & (1)(-2) \\ -1 & 2 \end{pmatrix}$$

$$\begin{pmatrix} -5 & 0 \\ -1 & 2 \end{pmatrix} \quad \begin{matrix} -1/5R1 \to R1 \\ \frac{1}{2}R2 \to R2 \end{matrix} \quad L = \begin{pmatrix} 1 & 0 \\ -1/2 & 1 \end{pmatrix}$$

$$A_2 = \begin{pmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{pmatrix}$$

$$\xrightarrow{\frac{1}{2}R1 + R2 \to R2}$$

$$\begin{pmatrix} 2(1/2) & -1(1/2) & 0(1/2) \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{pmatrix}$$

3

$$\begin{pmatrix} 2 & -1 & 0 \\ 0 & 3/2 & -1 \\ 0 & -1 & 2 \end{pmatrix} \xrightarrow{\frac{2}{3}R2 + R3 \to R3}$$

$$\begin{pmatrix} 2 & -1 & 0 \\ 0 & 3/2 & -1/2 \\ 0 & 0 & 4/3 \end{pmatrix} \Rightarrow U$$

$$A_2 = \begin{pmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{pmatrix}$$

$\xrightarrow{1/2R3 + R2 \to R2}$

$$\begin{pmatrix} 2 & -1 & 0 \\ -1 & 3/2 & 0 \\ 0 & -1 & 2 \end{pmatrix} \xrightarrow{\frac{2}{3}R2 + R1 \to R1}$$

$$\begin{pmatrix} 4/3 & 0 & 0 \\ -1 & 3/2 & 0 \\ 0 & -1 & 2 \end{pmatrix} \begin{matrix} \frac{2}{3}R2 \to R2 \\ \frac{3}{4}R1 \to R1 \\ 1/2R3 \to R3 \end{matrix} \xrightarrow{}$$

$$\begin{pmatrix} 1 & 0 & 0 \\ 2/3 & 1 & 0 \\ 0 & -1/2 & 1 \end{pmatrix} = L$$

L-matrix)

$$A_3 = \begin{pmatrix} 3 & 0 & 0 \\ -1 & 2 & 0 \\ 2 & 1 & 2 \end{pmatrix}$$ In this form!

~~A(6,7)~~ ✓

* for this case normalize diagonal and any underneath do a_{ij}/a_{ii} for bottom entries

$$L = \begin{pmatrix} 1 & 0 & 0 \\ -1/3 & 1 & 0 \\ 2/3 & 1/2 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ -1/3 & 1 & 0 \\ 2/3 & 1/2 & 0 \end{pmatrix}$$

~~Upper matrix~~:

$$A_3 = \begin{pmatrix} 3 & 0 & 0 \\ -1 & 2 & 0 \\ 2 & 1 & 2 \end{pmatrix} \xrightarrow{\frac{1}{3}R1 + R2 \to R2}$$

$$\begin{pmatrix} 3 & 0 & 0 \\ 0 & 2 & 0 \\ 2 & 1 & 2 \end{pmatrix} \xrightarrow{-\frac{2}{3}R1 + R3 \to R3}$$

$$\begin{pmatrix} 3 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 1 & 2 \end{pmatrix} \xrightarrow{-\frac{1}{2}R2 + R3 \to R3}$$

$$\begin{pmatrix} 3 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix} = U$$